

Generating Music w/ Code

(for non-musicians)

AMBIENT 1
MUSIC FOR AIRPORTS
BRIAN ENO

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"Music for Airports, at least one of the pieces on there, is structurally very, very simple. There are sung notes, sung by three women and myself. One of the notes repeats every $23 \frac{1}{2}$ seconds. It is in fact a long loop running around a series of tubular aluminum chairs in Conny Plank's studio. The next lowest loop repeats every $25 \frac{7}{8}$ seconds or something like that. The third one every $29 \frac{15}{16}$ seconds or something. What I mean is they all repeat in cycles that are called incommensurable – they are not likely to come back into sync again."

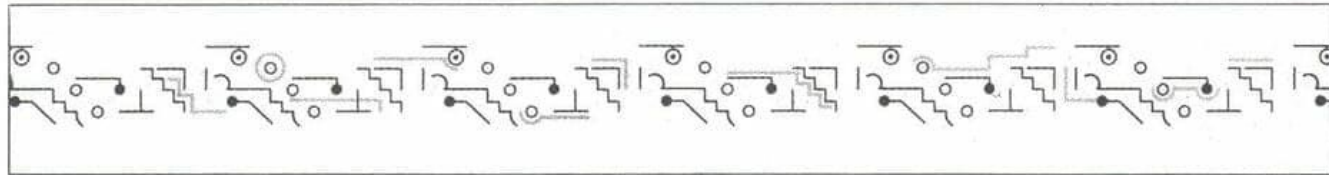


"I just set all of these loops running and let them configure in whichever way they wanted to, and in fact the result is very, very nice. The interesting thing is that it doesn't sound at all mechanical or mathematical as you would imagine. It sounds like some guy is sitting there playing the piano with quite intense feeling. The spacing and dynamics of "his" playing sound very well organized. That was an example of hardly interfering at all."



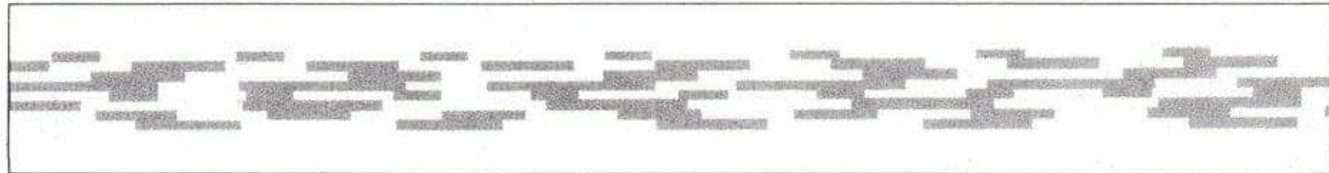
1/1

16·39



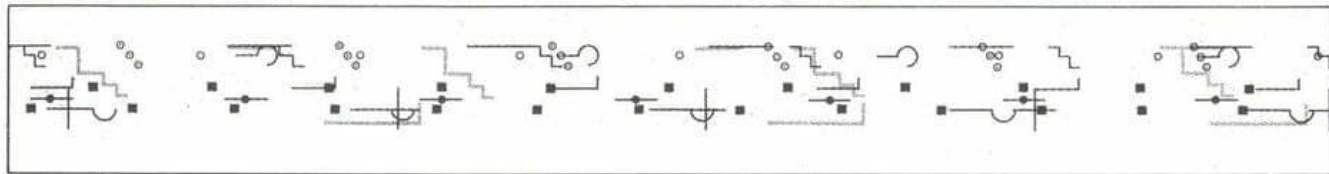
2/1

8·25



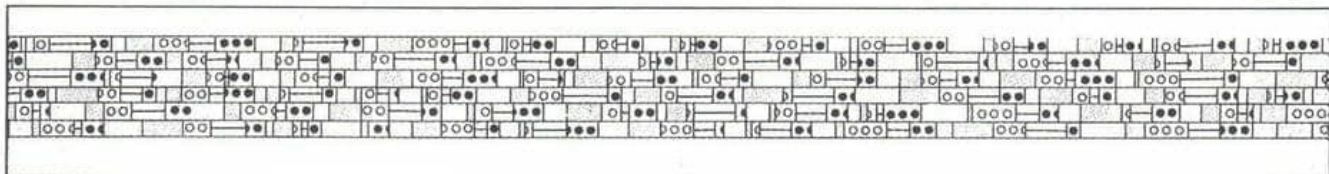
1/2

11·36



2/2

9·38





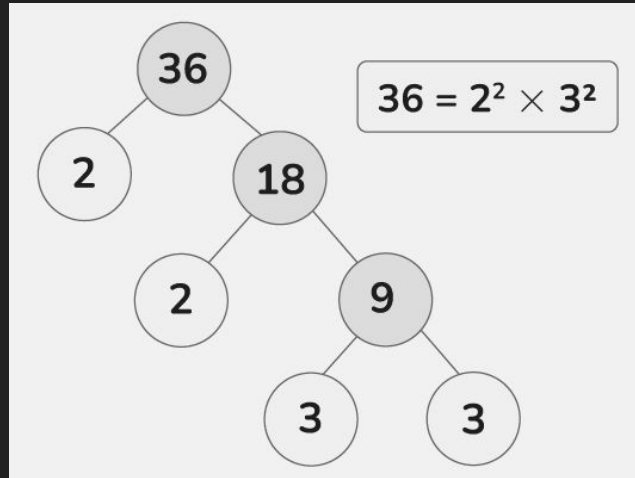
MATH

Fundamental Theorem of Arithmetic

Every integer is prime or can be represented uniquely as a product of prime numbers.

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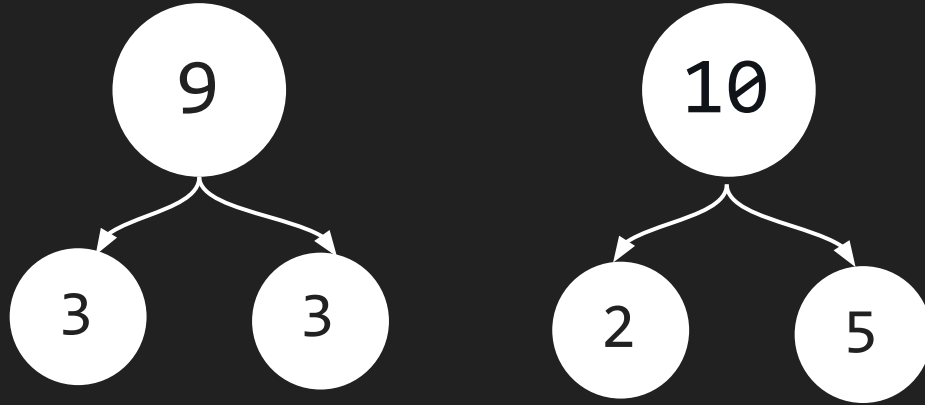


Coprime / Relatively Prime

Two integers A and B are "coprime" if the only positive integer that divides both numbers evenly is 1.

Coprime / Relatively Prime

Two or more integers are "coprime" if they don't have any prime factors in common.



9

10

$$\begin{array}{r} 9 \\ \hline 10 \end{array}$$



$$\begin{array}{r} 9 \\ \hline 10 \end{array}$$

9 notches



10 notches

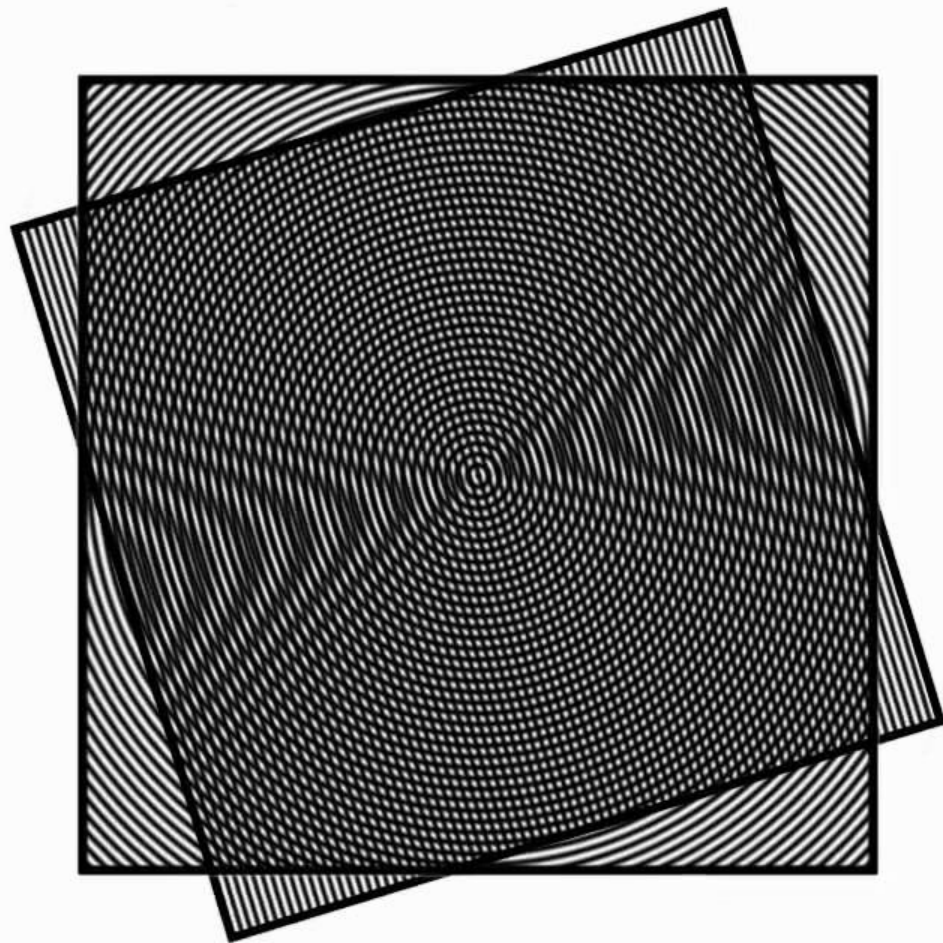
9 notches

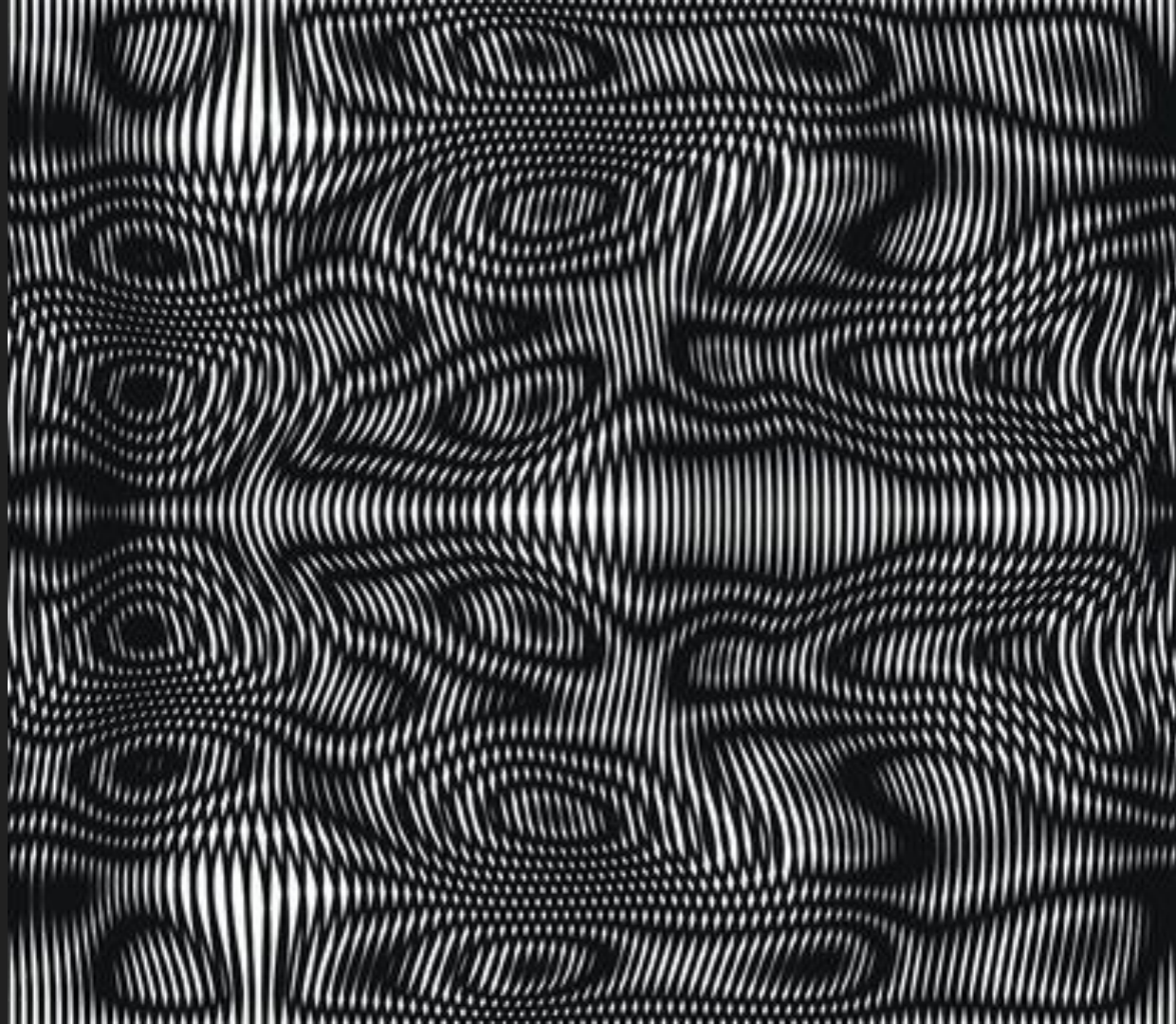


12 notches

Cycles with coprime ratios can create

PSEUDORANDOMNESS





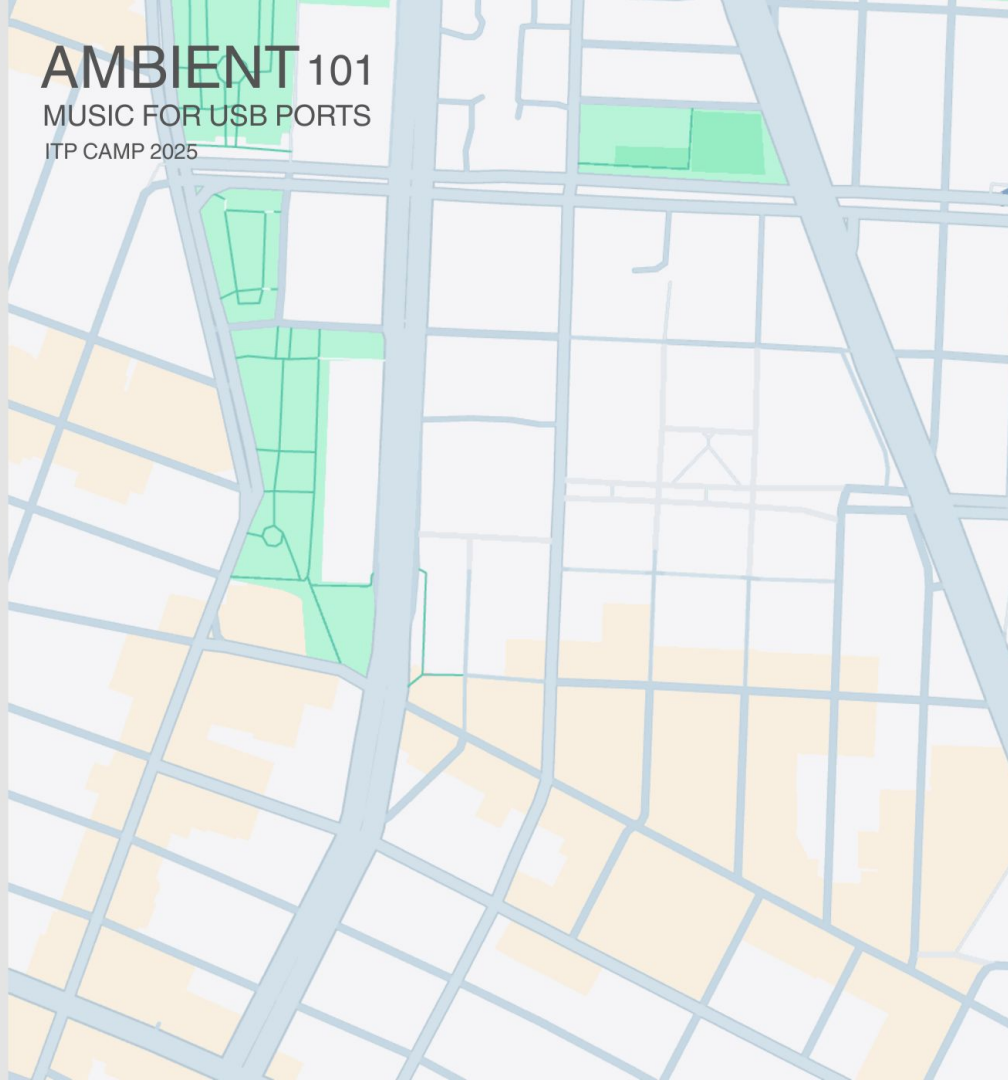
AMBIENT 1
MUSIC FOR AIRPORTS
BRIAN ENO

T

AMBIENT 101

MUSIC FOR USB PORTS

ITP CAMP 2025



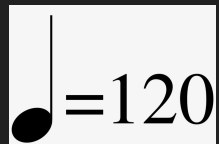
Workshop 1: Frequencies

[p5 sketch](#)

MUSIC THEORY

Time

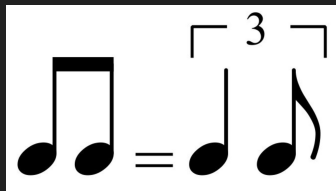
Tempo (BPM)



Meter

$\frac{5}{4}$ $\frac{2}{2}$ $\frac{7}{8}$ $\frac{2}{4}$ $\frac{3}{2}$ $\frac{4}{4}$

Groove /
Swing

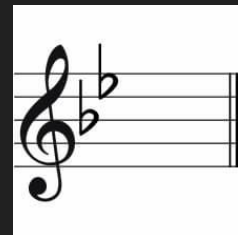


Space

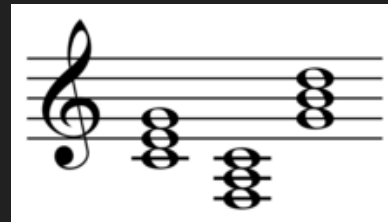
Pitch (Hz)



Key/Tonality



Harmony



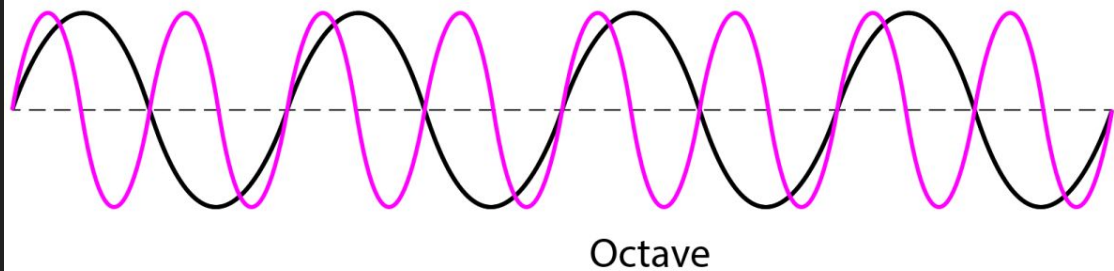
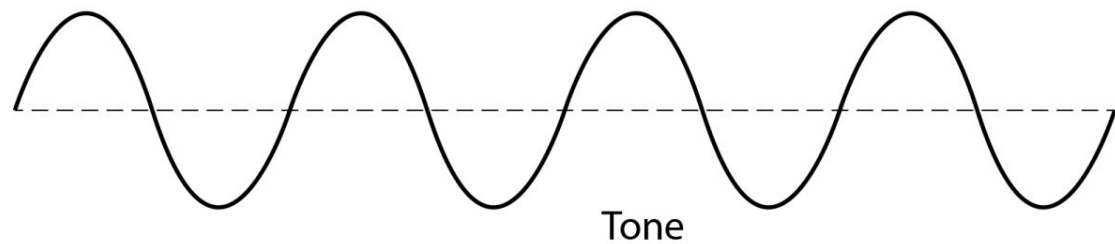
TIME = SPACE

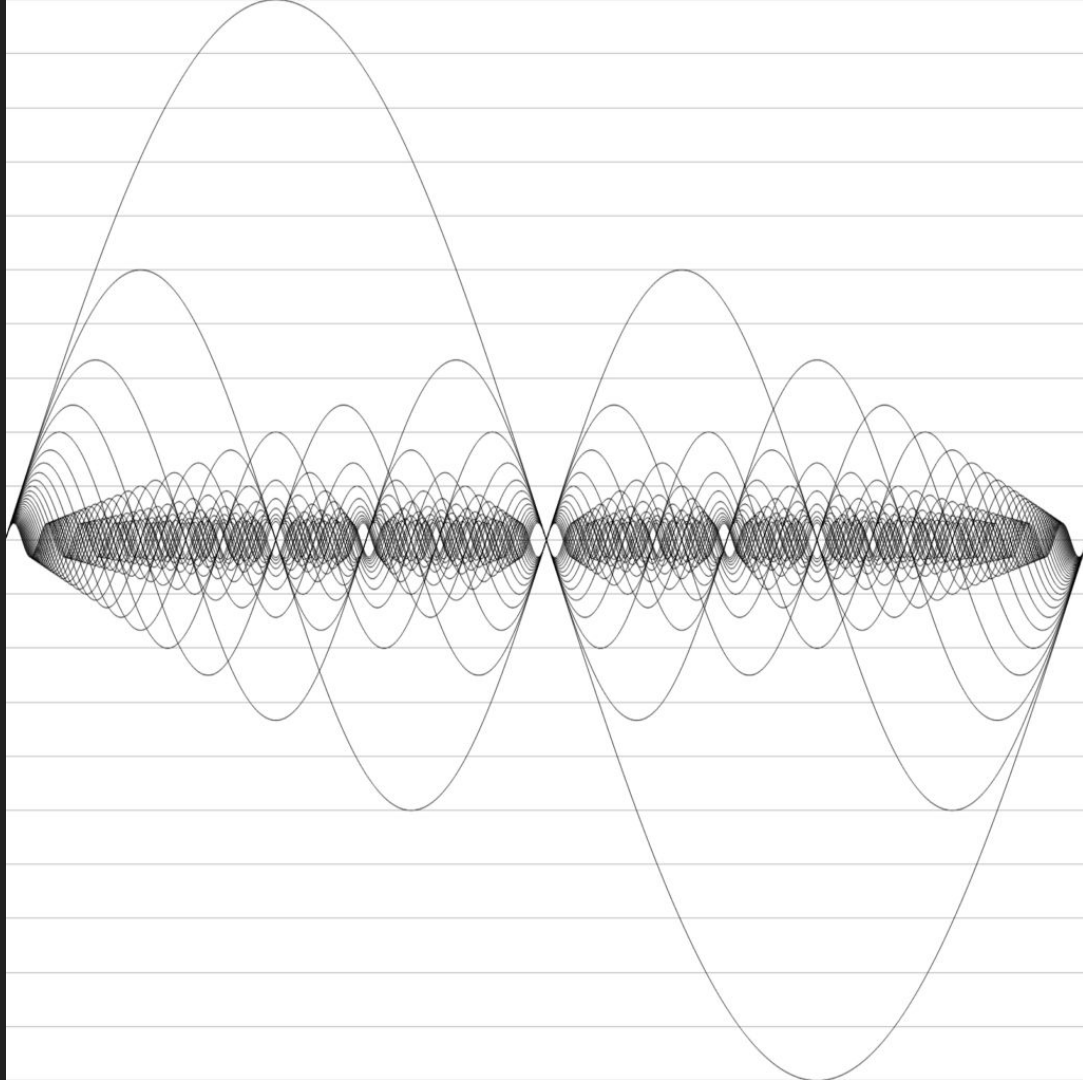
Rhythm = Pitch

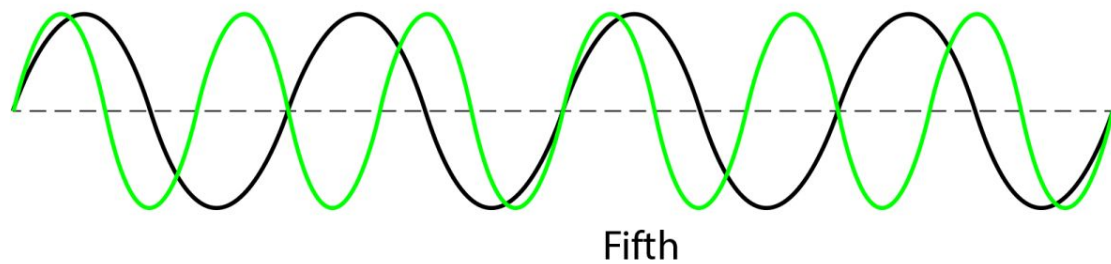
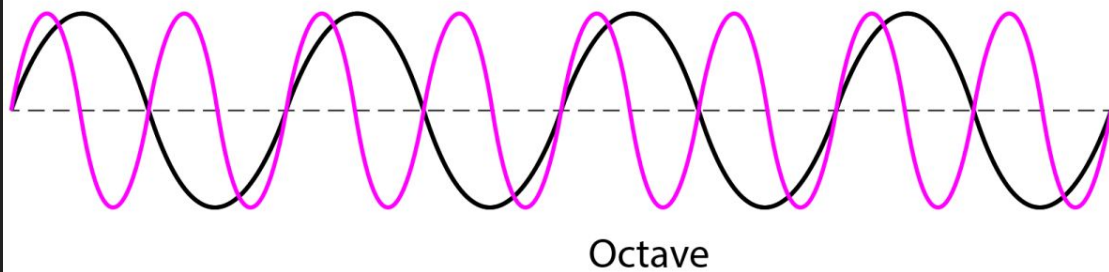
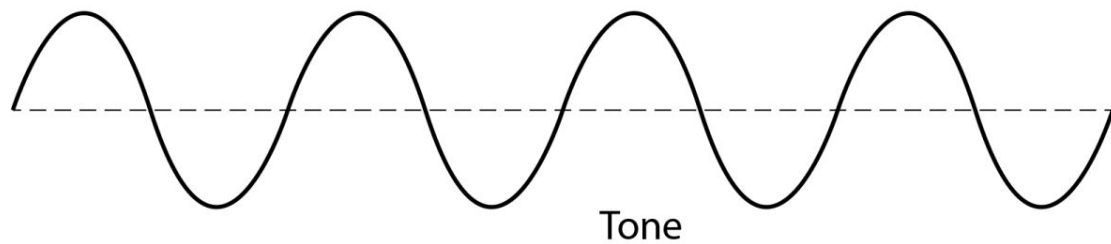
BPM = Hz

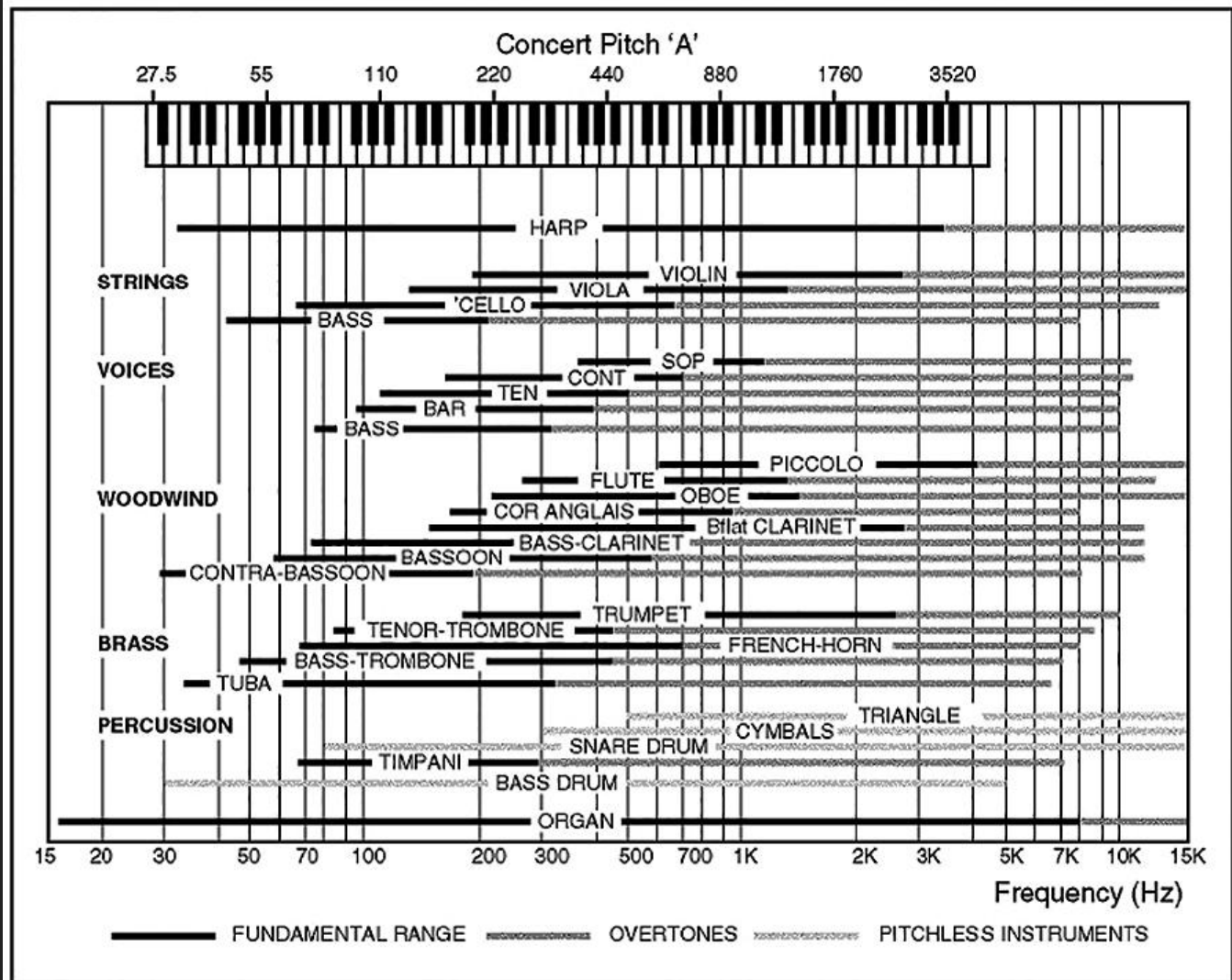
"how often?" = "how often?"













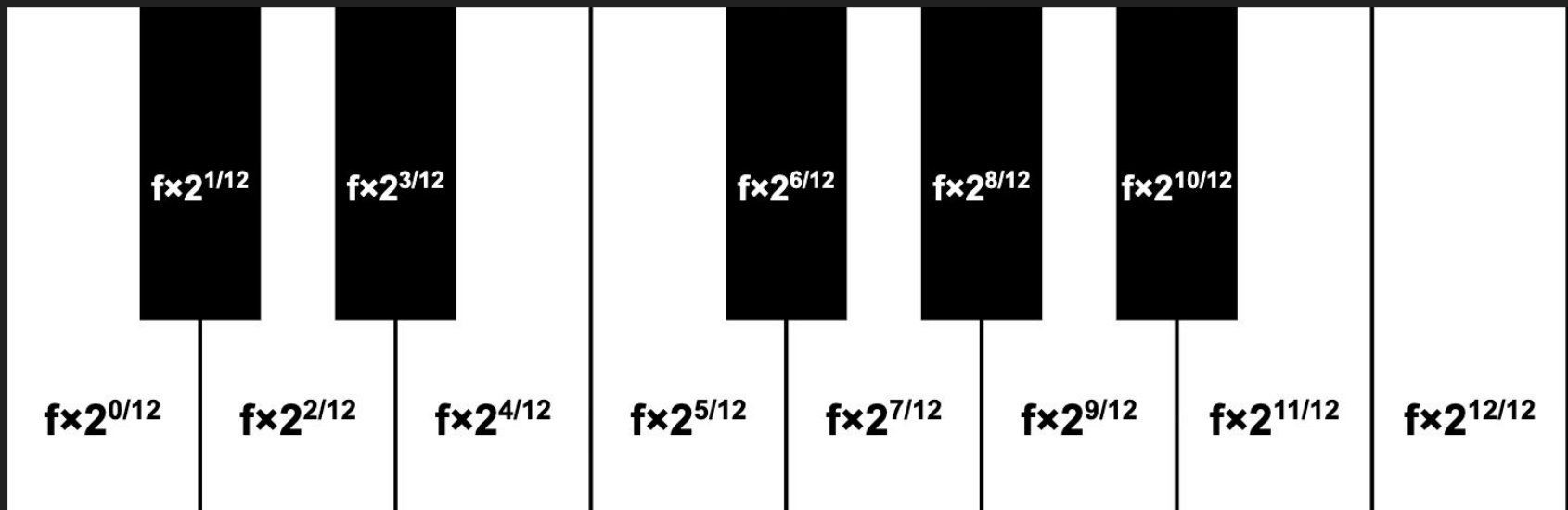
Workshop 2: Ratios

Tuning

MATH

12-TET

Twelve-tone Equal Temperament

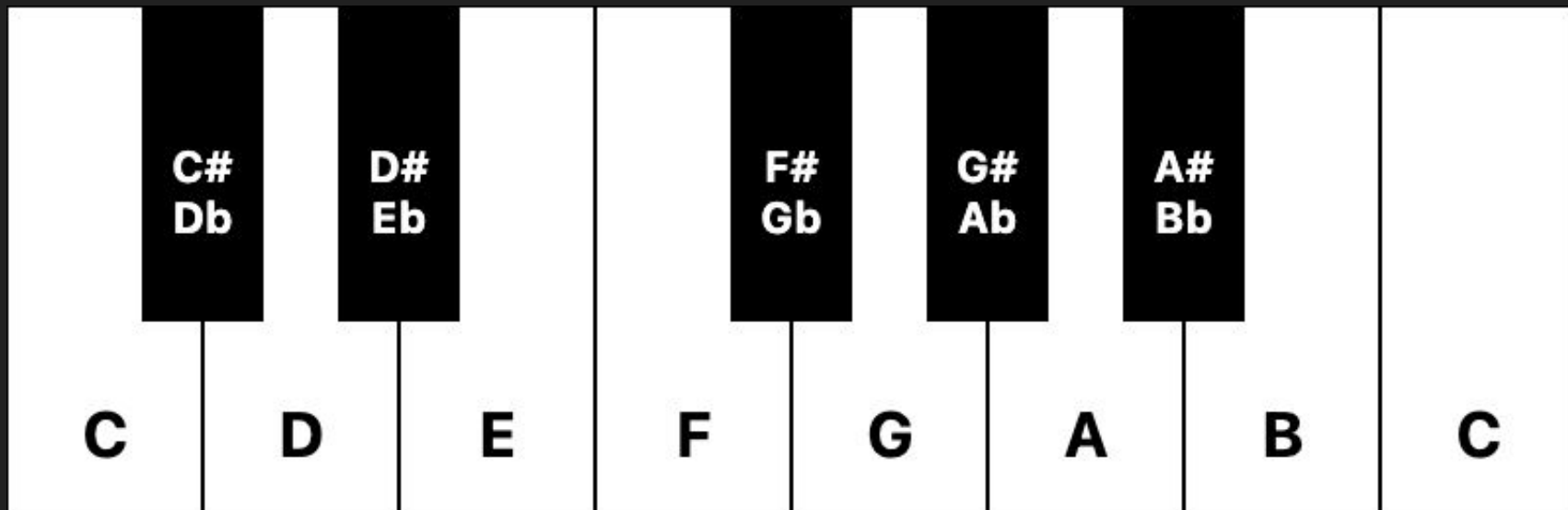


$$2^{(7 / 12)} =$$

1.49830707688

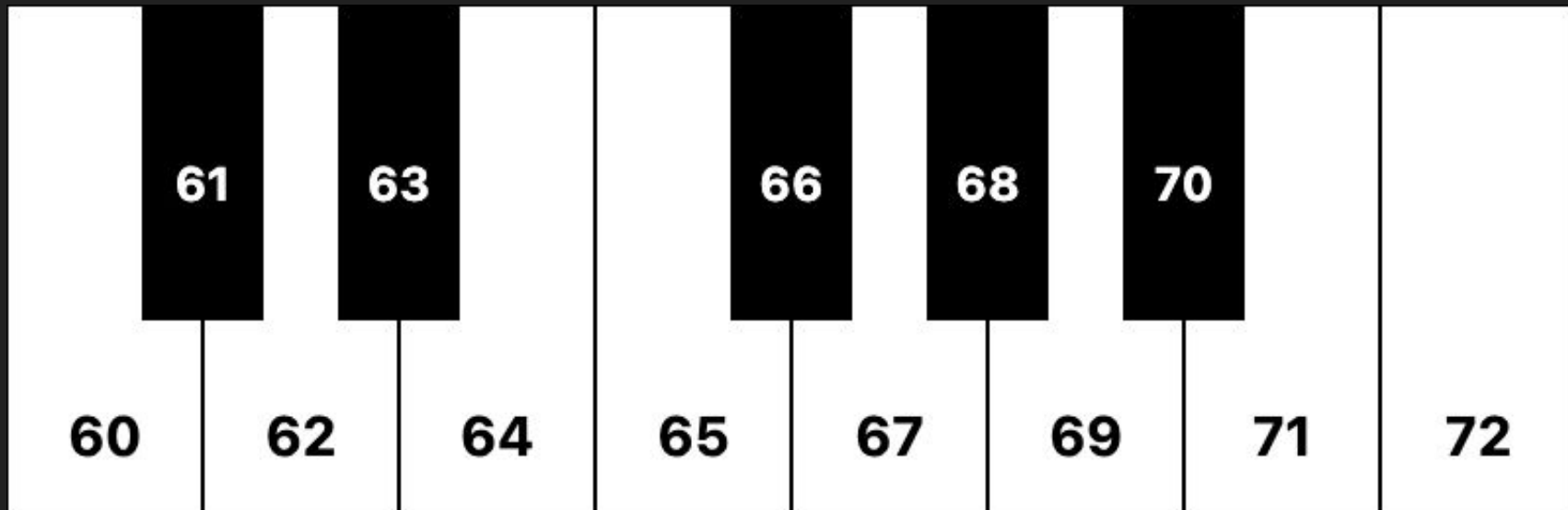
12-TET

Twelve-tone Equal Temperament



12-TET

Twelve-tone Equal Temperament



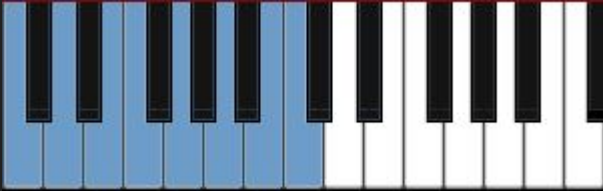
Workshop 3:

Chromatic Notes

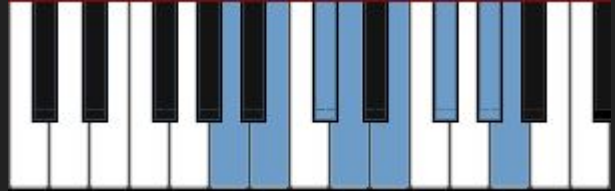
Scales & Modes

Key = tonic + scale

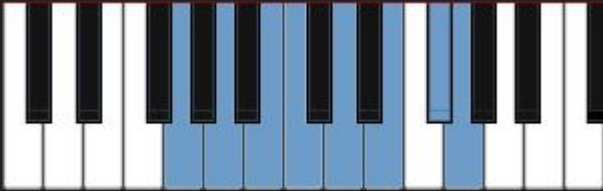
C Major



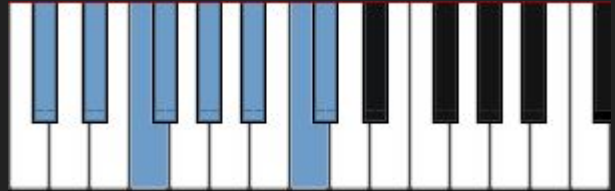
A Major



G Major

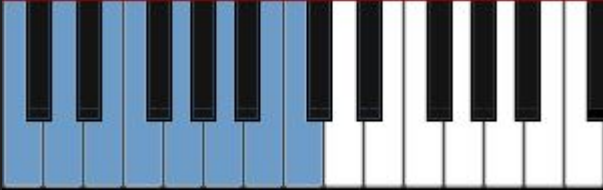


C# Major

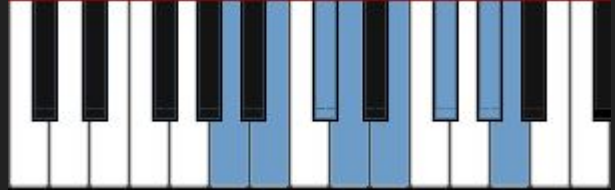


Major scale = WWHWWWH

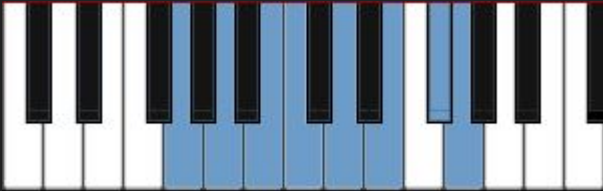
C Major



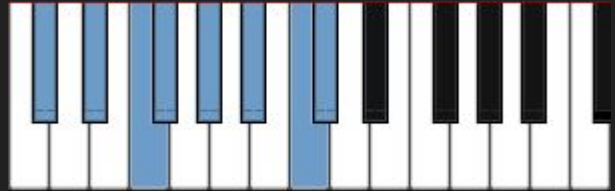
A Major

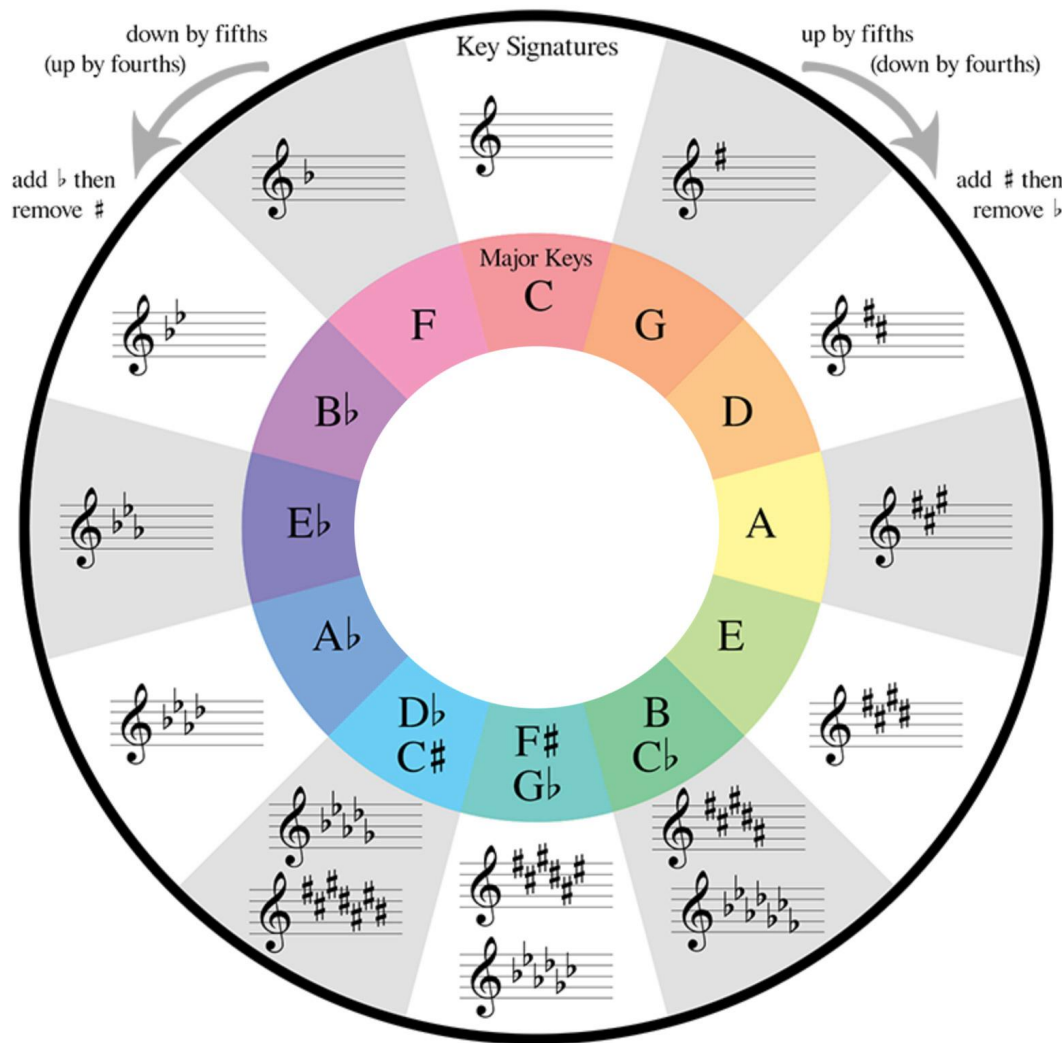


G Major



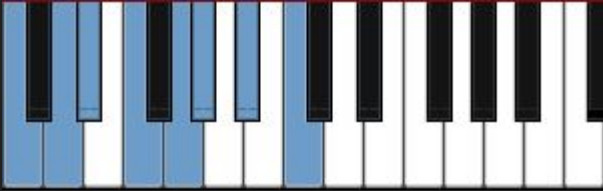
C# Major





Minor scale = WHWHWW

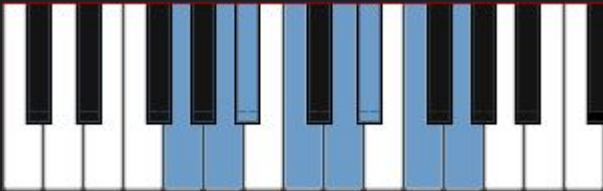
C Minor



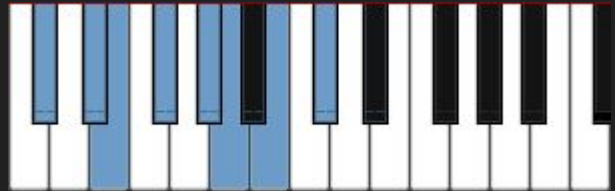
A Minor



G Minor

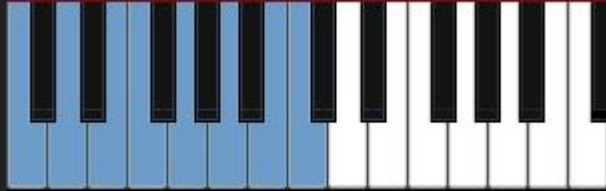


C# Minor



Relative Keys

C Major



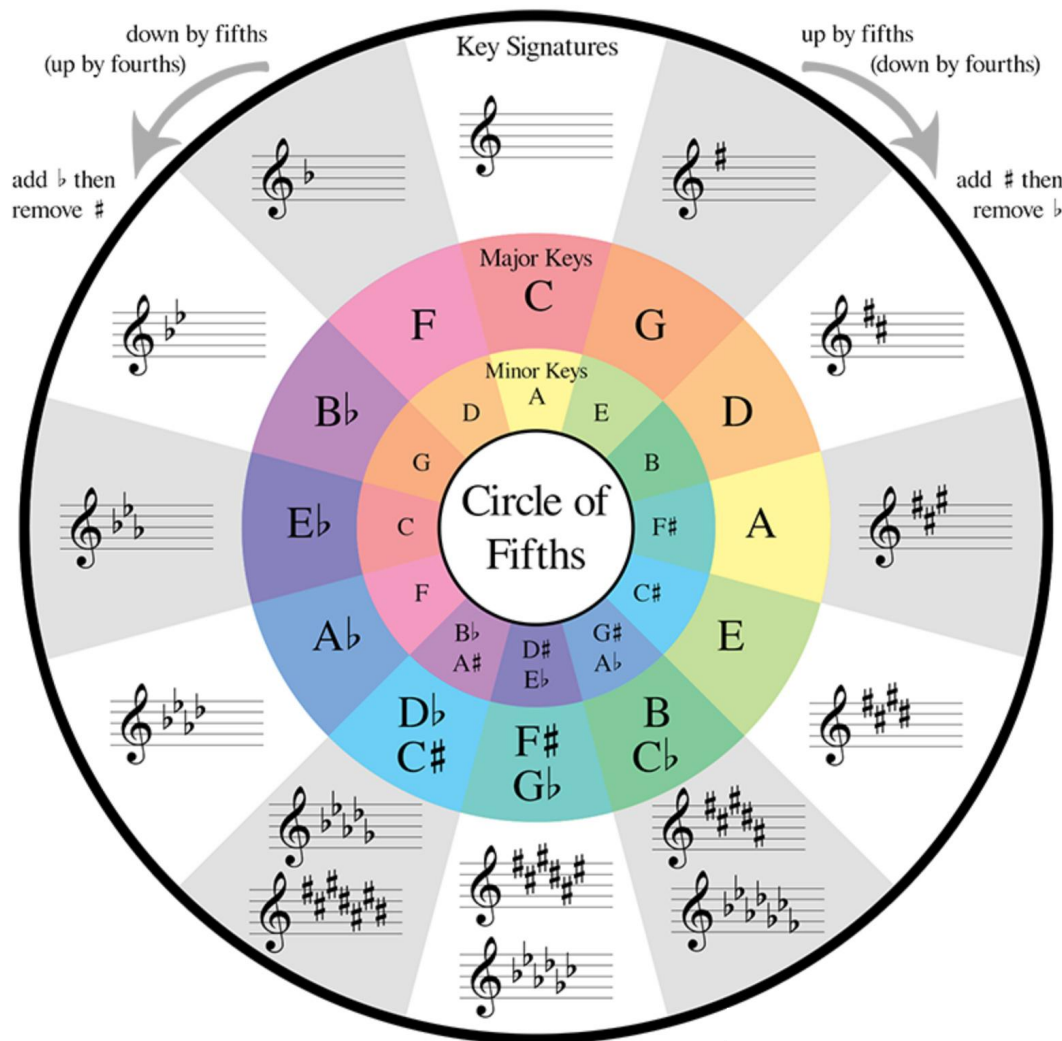
W W H W W W H



A Minor



W H W W H W W





Workshop 3:

Scales

Euclidean Rhythms

[The Euclidean Algorithm Generates Traditional Musical Rhythms \(2005\)](#)

[Euclidean Rhythm Generator](#)

Workshop 3:

Euclidean Rhythms

Markov Chains

"Mark of Markov"

When the piece was finished I listened to it and there was just one piano note I didn't like. It seemed to appear in the wrong place, so I simply edited it out. A lot of the so-called systems composers have this thing that the system is always right. You don't fiddle with it at all. Well, I don't think like that. I think that the system is as right as you judge it to be. If for some reason you don't like a bit of it you must trust your intuition on that. I don't take a doctrinaire approach to systems.





ROULETTE PRESENTS

BANG ON A CAN: LONG PLAY

BRIAN ENO MUSIC FOR AIRPORTS
PERFORMED BY BANG ON A CAN ALL-STARS
AND THE CHOIR OF TRINITY WALL STREET

